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Student ID	668596
Course	DSA 3050A — Business Intelligence & Data Visualization
Assessment	End-Semester Practical Examination
Dataset Used	Superstore Dataset
Data Source	Kaggle – https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
Submission Date	April 18, 2026
Total Marks	100 Marks

Part A: Data Acquisition and Understanding (10 Marks)

1. Domain and Business Problem

Domain: Retail / Sales Analytics

This project is developed within the retail analytics domain, focusing on analyzing sales transactions, customer behavior, and product performance across different regions and time periods.

Business Problem:

Retail businesses generate large volumes of transactional data, but without proper analytical tools, it becomes difficult to extract meaningful insights. The organization lacks an efficient system to monitor key performance indicators such as sales revenue, profit margins, customer segments, and regional performance.

As a result, decision-makers struggle to:

- identify high-performing and underperforming products
- analyze trends over time
- compare performance across regions
- understand customer purchasing behavior

This project aims to solve these challenges by developing an interactive Power BI dashboard that provides insights into sales performance, profitability, and trends. The solution enables stakeholders to make data-driven decisions, improve business strategies, and optimize performance.

2. Dataset Source

The dataset used in this project is the **Superstore Sales Dataset**, obtained from a public data repository (Kaggle).

The dataset contains real-world simulated retail transaction data and is widely used for business intelligence and data analytics practice.

It was downloaded in CSV format and imported into Power BI for further processing, including:

- data cleaning
- transformation

- modeling
- visualization

The dataset represents sales transactions across multiple regions, product categories, and customer segments over time.

Superstore Dataset

1138 | Code | Download

Data Card | Code (302) | Discussion (3) | Suggestions (0)

Sample - Superstore.csv (2.29 MB)

Detail | **Compact** | Column | 10 of 21 columns

Row ID	Order ID	Order Date	Ship Date	Ship Mode
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class

Data Explorer
Version 1 (2.29 MB)

Sample - Superstore.csv

Summary
1 file
21 columns

3. Key Tables and Fields

The dataset initially exists as a single table but is transformed into a **relational model** consisting of a fact table and multiple dimension tables.

Fact Table: Orders (Main Transaction Table)

This is the central table containing all transactional data and numerical measures.

Key fields include:

- **Order ID** – unique identifier for each order
- **Order Date** – date when the order was placed
- **Ship Date** – date when the order was shipped
- **Customer ID** – unique identifier for customers
- **Customer Name** – name of the customer
- **Segment** – customer category (Consumer, Corporate, Home Office)
- **Country, City, State, Region** – geographical details
- **Product Name** – name of the product sold

- **Category** – product category (e.g., Furniture, Technology, Office Supplies)
- **Sub-Category** – detailed product classification
- **Sales** – revenue generated from the transaction
- **Quantity** – number of units sold
- **Discount** – discount applied
- **Profit** – profit earned from the transaction

Dimension Tables

To improve analysis and modeling, the dataset is split into the following dimension tables:

Customer Dimension (DimCustomer)

- Customer ID
- Customer Name
- Segment

Purpose:

Used to analyze customer behavior, segmentation, and purchasing patterns.

Product Dimension (DimProduct)

- Product Name
- Category
- Sub-Category

Purpose:

Used to analyze product performance and category-level trends.

Date Dimension (DimDate)

- Date
- Year
- Month
- Quarter

Purpose:

Used for time-based analysis such as trends, seasonality, and growth.

4. Suitability of the Dataset for Advanced Power BI Analysis

The Superstore dataset is highly suitable for advanced Power BI analysis due to the following reasons:

1. Adequate Data Volume

The dataset contains over 9,000 records, which satisfies the requirement for large-scale analysis and ensures meaningful insights can be generated.

2. Variety of Data Types

The dataset includes:

- **Numerical data** (Sales, Profit, Quantity, Discount)

Sales – total revenue generated per transaction

Profit – profit earned from each sale

Quantity – number of items sold

Discount – discount applied to the transaction

- **Categorical data** (Category, Segment, Region)

Category and Sub-Category – classify products

Segment – defines customer groups (Consumer, Corporate, Home Office)

Region, State, City – represent geographical locations

This allows for diverse analytical operations such as aggregation, comparison, and segmentation.

3. Time Intelligence Capability

The presence of date fields (Order Date and Ship Date) enables:

- trend analysis over time
- year-to-date calculations

- seasonal pattern identification

4. Supports Star Schema Modeling

The dataset can be structured into:

- a central **fact table (Orders)**
- multiple **dimension tables (Customer, Product, Date)**

This supports the implementation of a **star schema**, which improves query performance and simplifies reporting.

5. Enables Advanced DAX Calculations

The dataset supports the creation of advanced measures such as:

- total revenue and profit
- growth rates
- ranking of products and categories
- time-based calculations (YTD, previous period analysis)

6. Real-World Business Relevance

The dataset reflects realistic retail operations, making it suitable for simulating real-world business intelligence scenarios. It allows the development of dashboards that can support decision-making in areas such as sales strategy, customer targeting, and operational efficiency.

Screenshot of the raw dataset

Superstore Dataset.csv - Excel (Product Activation Failed)

	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit		
1	#####	Second Class	CG-12520	Claire Gutierrez	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10	Furniture Bookcases	Bush Somerset	261.96	2	0	41.9136		
2	#####	Second Class	CG-12520	Claire Gutierrez	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10	Furniture Chairs	Honolulu Deluxe	731.94	3	0	219.582		
3	#####	Second Class	DV-13045	Darrin Varner	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10	Office Supplies Labels	Self-Adhesive	14.62	2	0	6.8714		
4	#####	Standard	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10	Furniture Tables	Bretford Classic	957.5775	5	0.45	-383.031		
5	#####	Standard	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10	Office Supplies Storage	Eldon Folio	22.368	2	0.2	2.5164		
6	#####	Standard	BH-11710	Brosina Hirsch	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10	Furniture Furnishings	Eldon Express	48.86	7	0	14.1694		
7	#####	Standard	BH-11710	Brosina Hirsch	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10	Office Supplies Art	Newell 32	7.28	4	0	1.9656		
8	#####	Standard	BH-11710	Brosina Hirsch	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10	Technology Phones	Mitel 532C	907.152	6	0.2	90.7152		
9	#####	Standard	BH-11710	Brosina Hirsch	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10	Office Supplies Binders	DXL Angle	18.504	3	0.2	5.7825		
10	#####	Standard	BH-11710	Brosina Hirsch	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10	Office Supplies Appliances	Belkin F5C	114.9	5	0	34.47		
11	#####	Standard	BH-11710	Brosina Hirsch	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10	Furniture Tables	Chromcraft	1706.184	9	0.2	85.3092		
12	#####	Standard	BH-11710	Brosina Hirsch	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10	Technology Phones	Konftel 25	911.424	4	0.2	68.3568		
13	#####	Standard	AA-10480	Andrew A.	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10	Office Supplies Paper	Xerox 196	15.552	3	0.2	5.4432		
14	#####	Standard	IM-15070	Irene Madson	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10	Office Supplies Binders	Fellowes F	407.976	3	0.2	132.5922		
15	#####	Standard	HP-14815	Harold Palmer	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10	Office Supplies Appliances	Holmes Re	68.81	5	0.8	-123.858		
16	#####	Standard	HP-14815	Harold Palmer	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10	Office Supplies Binders	Storex Dui	2.544	3	0.8	-3.816		
17	#####	Standard	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10	Office Supplies Storage	Stur-D-Sto	665.88	6	0	13.3176		
18	#####	Second Class	AG-10270	Alejandro	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10	Office Supplies Storage	Fellowes S	55.5	2	0	9.99		
19	9/1/2014	Second Class	ZD-21925	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10	Office Supplies Art	Newell 34	8.56	2	0	2.4824		
20	9/1/2014	Second Class	ZD-21925	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10	Technology Phones	Cisco SPA	213.48	3	0.2	16.011		
21	9/1/2014	Second Class	ZD-21925	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10	Office Supplies Binders	Wilson Jor	22.72	4	0.2	7.384		
22	#####	Standard	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10	Office Supplies Art	Newell 31	19.46	7	0	5.0596		

Part B: Data Cleaning and Transformation in Power Query

(20 Marks)

Raw dataset in Power Query before cleaning

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [1] Superstore Dataset

Table.TransformColumnTypes(#"Promoted Headers",{{"Row ID", Int64.Type}, {"Order ID", type text}, {"Order Date", type date}, {"Ship Date", type date}, {"Ship Mode", type text}}

Row ID Order ID Order Date Ship Date Ship Mode

Valid 100% Error 0% Empty 0% Valid 100% Error 0% Empty 0% Valid 100% Error 0% Empty 0% Valid 100% Error 0% Empty 0% Valid 100% Error 0% Empty 0%

1000 distinct, 1000 unique 479 distinct, 226 unique 373 distinct, 133 unique 398 distinct, 152 unique 4 distinct, 0 unique

1	CA-2016-152156	11/8/2016	11/11/2016	Second Class
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class

21 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 3:27 PM

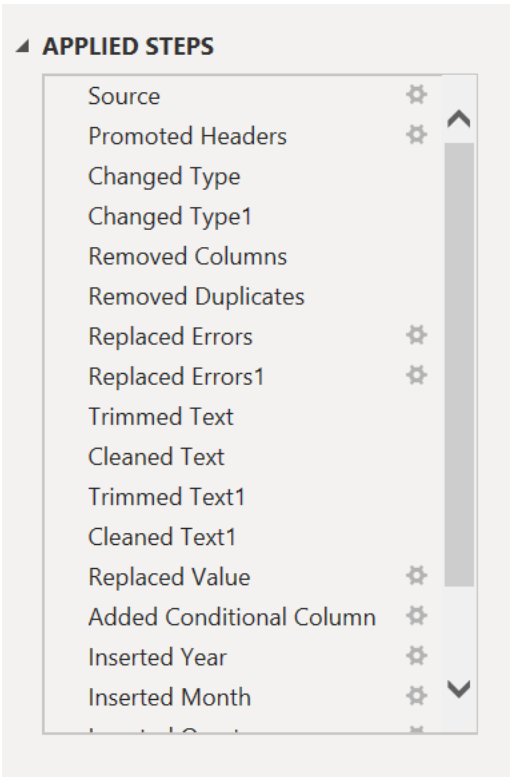
Query Settings

PROPERTIES Name Superstore Dataset All Properties

APPLIED STEPS Source Promoted Headers Changed Type

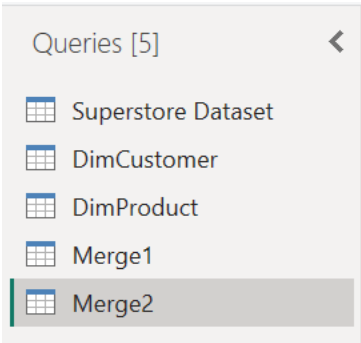
The dataset was initially loaded into Power Query in its raw form. At this stage, the data contained inconsistencies such as incorrect data types, unnecessary columns, and unprocessed text fields. The purpose of this step was to inspect the dataset and identify areas that required cleaning and transformation.

Transformation Steps pane



The Applied Steps pane shows all transformation steps performed on the dataset. Each step represents a cleaning or transformation operation such as changing data types, removing duplicates, filtering invalid records, and creating new columns. This ensures that the data preparation process is transparent and reproducible.

Merged Queries



Separate dimension tables for customers and products were created and merged into the main dataset using the Merge Queries function. The merging was performed using key fields such as Customer ID and Product Name. A left outer join was applied to retain all records from the main table while incorporating related information from the dimension tables. This process enhances data organization and supports relational modeling.

Explanations of transformations

- Data Type Correction

Data types were corrected for key fields such as dates and numerical values to ensure accurate calculations and proper time-based analysis.

- Removal of Unnecessary Columns

Irrelevant columns such as Row ID were removed to simplify the dataset and improve performance.

- Handling Missing Values

Missing values were identified and handled by either removing incomplete records or replacing them with appropriate values, ensuring data consistency.

- Removal of Duplicates

Duplicate records were removed to prevent double counting and ensure accurate aggregation of results.

- Text Cleaning (Trim & Clean)

Text fields were cleaned to remove extra spaces and non-printable characters, improving consistency in categorical data.

- Standardization of Categories

Category values were standardized to ensure uniform grouping and accurate comparisons.

- Conditional Column Creation

A new column was created to classify transactions as profit or loss, enabling easier identification of performance patterns.

- Date Transformation

Date components such as year, month, and quarter were extracted to support time-series analysis

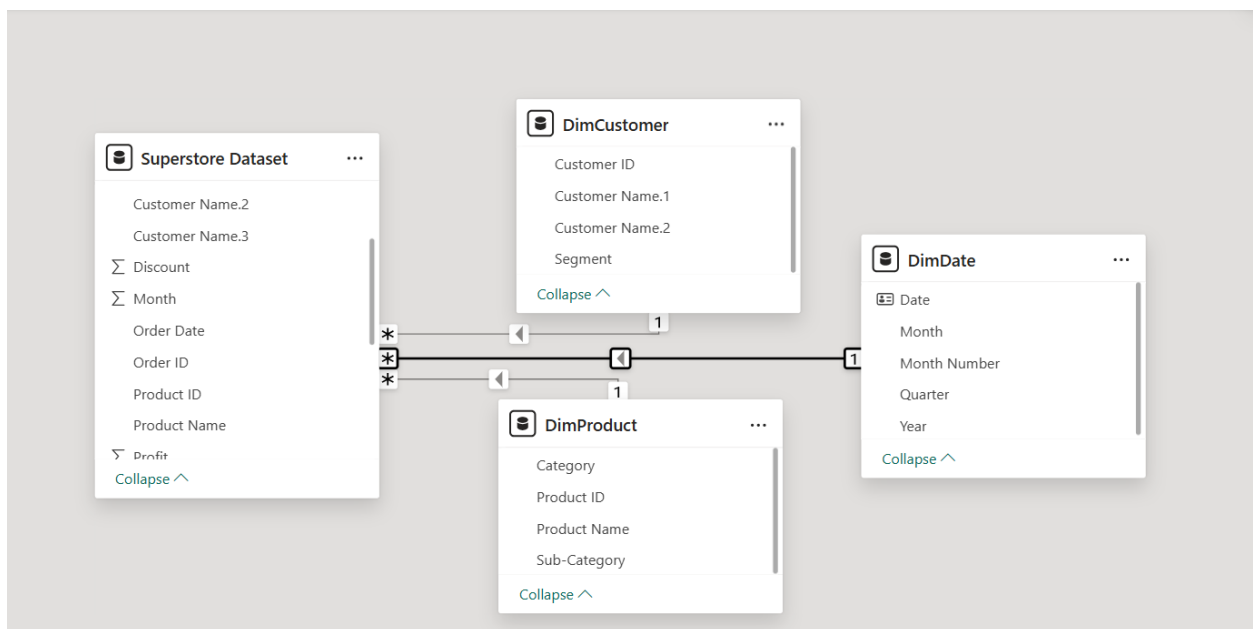
- Column Splitting

Columns such as Customer Name were split into multiple fields to improve data structure and enable more detailed analysis.

Part C: Data Modeling (15 Marks)

This section is about turning your cleaned data into a proper analytical model. The exam requires relationships, fact and dimension table identification, correct cardinality, a Date table, and an explanation of why the structure supports efficient reporting.

Model view



- Fact and Dimension Tables

Fact Table

The **Superstore Dataset** was used as the fact table. It contains transactional data where each row represents a sales record. It includes measurable values such as Sales, Profit, Quantity, and Discount, which are used for analysis.

Dimension Tables

The **DimCustomer** table was created to store customer-related information such as Customer ID, customer name, and segment. It is used to analyze sales by customer characteristics.

The **DimProduct** table contains product-related attributes such as Product ID, Product Name, Category, and Sub-Category. It enables product-level analysis.

The **DimDate** table was created using DAX to support time-based analysis. It includes fields such as Date, Year, Month, Month Number, and Quarter.

3. Relationships

Relationships were established between the fact table and dimension tables using key fields. The **Customer ID** field connects the DimCustomer table to the Superstore Dataset. The **Product ID** field connects the DimProduct table to the Superstore Dataset. The **Date** field from the DimDate table is linked to the **Order Date** field in the Superstore Dataset.

4. Cardinality and Cross-Filter Direction

All relationships were configured using **one-to-many (1:*) cardinality**, where each dimension table contains unique values that relate to multiple records in the fact table.

The **cross-filter direction** was set to **single**, allowing filtering to flow from dimension tables to the fact table. This ensures better performance and avoids ambiguity in the model.

5. Date Table Creation

A dedicated Date table (DimDate) was created using DAX to enable time intelligence analysis. The table was generated using the minimum and maximum order dates from the dataset and includes additional attributes such as Year, Month, and Quarter. The table was then marked as a date table and related to the fact table.

6. Why the Model Supports Efficient Reporting

The star schema structure improves performance by reducing redundancy and simplifying relationships. It allows users to analyze data efficiently by filtering across different dimensions such as customer, product, and time. This structure also supports accurate DAX calculations and enhances report usability.

Part D: DAX Measures and Calculated Columns (20 Marks)

DAX measures and calculated columns were created in Power BI to support advanced analysis, dashboard interactivity, and business decision-making. Calculated columns were used for row-level categorization, while measures were used to calculate aggregated results dynamically based on the reporting context.

Calculated Columns

Profit Status

Profit Status	
profit	
profit	
profit	
profit	
loss	
profit	
loss	
profit	
profit	
profit	
profit	
profit	
profit	
profit	
profit	
profit	
profit	
profit	
profit	
profit	
loss	

The Profit Status column was created during the data transformation stage to classify transactions as either profit or loss based on the Profit value.

Sales Band

```
1 Sales Band =  
2 SWITCH(  
3     TRUE(),  
4     'Superstore Dataset'[Sales] < 100, "Low Sales",  
5     'Superstore Dataset'[Sales] < 500, "Medium Sales",  
6     "High Sales"  
7 )
```

The Sales Band column was created using DAX to group transactions into low, medium, and high sales categories. This supports segmentation and analysis.

DAX Measures

1. Total revenue

```
1 Total Revenue = SUM('Superstore Dataset'[Sales])
```

Sub-Category
Total Revenue
Year

The **Total Revenue** measure was created by summing the Sales column in the fact table. It calculates the total sales generated by the business and serves as a key indicator of overall business performance.

How it helps decision-making

This measure helps managers monitor total business income and compare sales performance across regions, products, and customer segments.

2. Gross profit/ Total profit

```
1 Total Profit = SUM('Superstore Dataset'[Profit])
```

Sub-Category
Total Profit
Total Revenue

The **Total Profit** measure sums the Profit column to calculate the total profit earned from all transactions.

How it helps decision making

This measure helps stakeholders evaluate whether sales activities are profitable and identify areas where profitability is improving or declining.

3. Profit Margin (%)

```
1 Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)
```

Profit
Profit Margin %
Profit Status

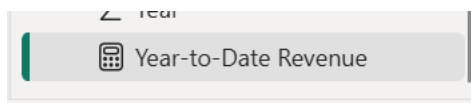
The **Profit Margin %** measure divides gross profit by total revenue to calculate the percentage of revenue retained as profit.

How it helps decision making

This measure helps decision-makers assess operational efficiency, pricing effectiveness, and the profitability of the business.

4. Year-to-Date Revenue

```
1 Year-to-Date Revenue = TOTALYTD([Total Revenue], 'DimDate'[Date])
```



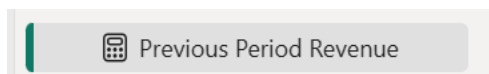
The **Year-to-Date Revenue** measure calculates cumulative revenue from the beginning of the year up to the current reporting period using the Date dimension table.

How it helps decision making

This measure helps stakeholders track annual progress and evaluate current performance against targets or previous periods.

5. Previous Period Revenue

```
1 Previous Period Revenue =  
2 CALCULATE(  
3 | [Total Revenue],  
4 | SAMEPERIODLASTYEAR('DimDate'[Date])  
5 )
```



The **Previous Period Revenue** measure calculates revenue for the same period in the previous year using the Date dimension table.

How it helps decision making

This measure enables period-to-period comparison and helps determine whether current performance is improving or declining.

6. Growth %

```
1 Growth % =  
2 DIVIDE(  
3     [Total Revenue] - [Previous Period Revenue],  
4     [Previous Period Revenue],  
5     0  
6 )
```

 Growth %

The **Growth %** measure compares current revenue with previous period revenue and expresses the change as a percentage.

How it helps decision making

This measure helps management identify whether the business is growing or declining over time and supports performance monitoring.

7. Top Category Rank

```
1 Top Category Rank =  
2 RANKX(  
3     ALL('DimProduct'[Category]),  
4     [Total Revenue],  
5     ,  
6     DESC  
7 )
```

 Top Category Rank

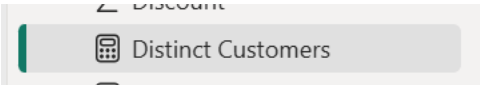
The **Top Category Rank** measure ranks product categories according to total revenue, with the highest-performing category receiving the top rank.

How it helps decision making

This measure helps identify the strongest categories and supports decisions on inventory, marketing, and resource allocation.

8. Distinct Customers

```
1 Distinct Customers = DISTINCTCOUNT('Superstore Dataset'[Customer ID])
```



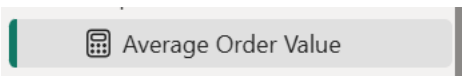
The **Distinct Customers** measure counts the number of unique customers based on Customer ID.

How it helps decision making

This measure helps the business understand customer reach and evaluate how many unique customers are being served.

9. Average Order Value

```
1 Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
```



The **Average Order Value** measure divides total revenue by total orders to calculate the average value generated per customer order.

How it helps decision making

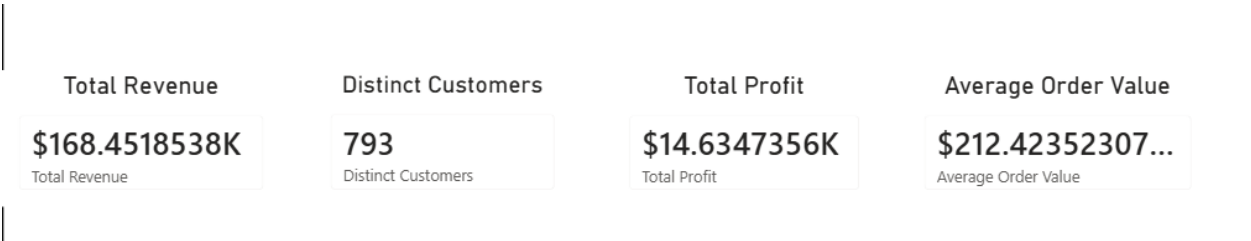
This measure helps management understand customer spending behavior and evaluate whether the business is generating high-value transactions.

- Several DAX measures were created in Power BI to support financial analysis, time intelligence, customer analysis, and category performance evaluation. These measures were stored in the fact table and formatted appropriately as currency, percentage, or whole numbers to improve interpretation in reports.

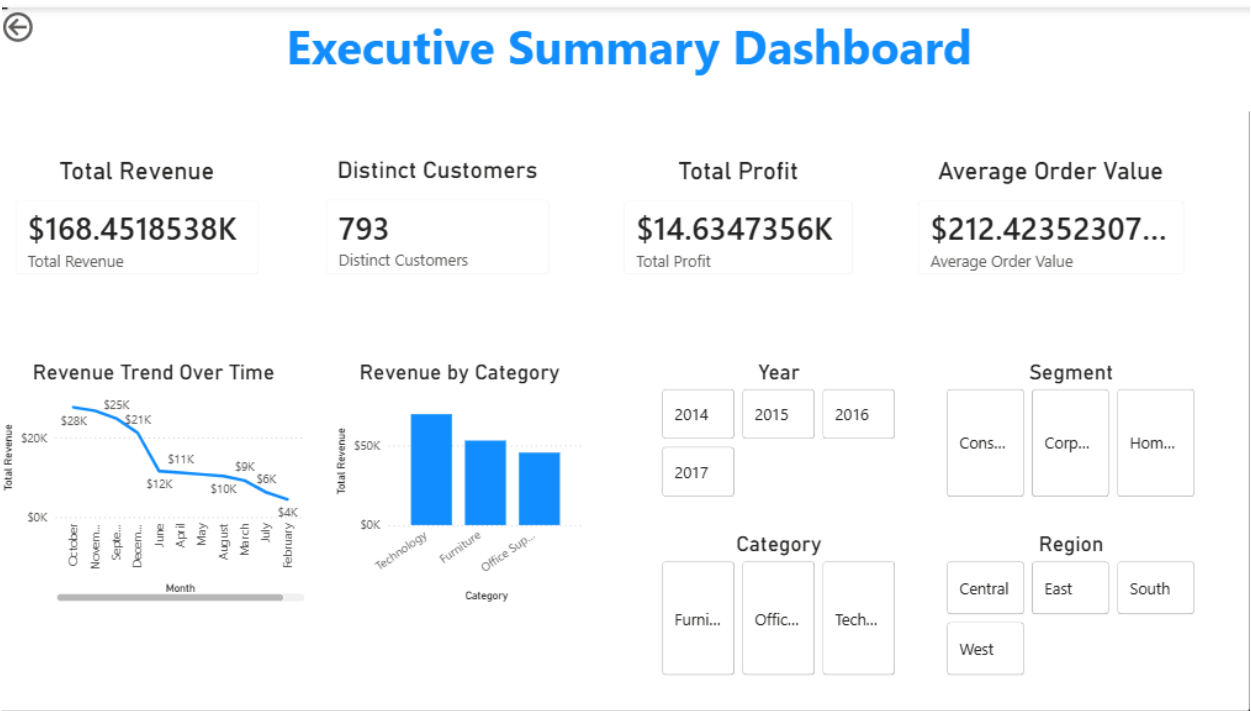
Part E: Dashboard Design and Visualization (20 Marks)

Page 1: Executive Summary

The first report page was designed as an executive summary to provide a high-level overview of business performance. It presents key metrics, major sales trends, category comparisons, and filtering options for quick decision-making.



KPI cards were used to present the most important summary metrics at a glance. These include total revenue, gross profit, distinct customers, and average order value. This design helps executives quickly assess business performance without needing detailed exploration.

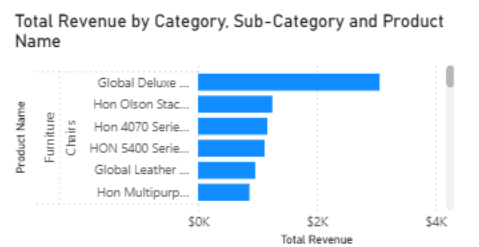


The Executive Summary dashboard provides a high-level overview of business performance using KPI cards, trend analysis, and category comparisons. KPI cards display key metrics such as total revenue, profit, and customer count. A line chart illustrates revenue trends over time, while a bar chart compares performance across product categories. Interactive slicers enable users to filter data by year, segment, category, and region, enhancing user-driven analysis. The layout was designed to be clean, logical, and business-friendly, ensuring quick insights for decision-makers.

Page 2: Detailed Analysis

The second report page was designed for deeper exploration of business performance. It includes drill-down analysis, tabular summaries, and breakdown visuals to help users investigate performance drivers.

Drill-down Visual



A drill-down visual was created using a product hierarchy of category, sub-category, and product name. This allows users to move from a high-level category view to more detailed product-level analysis.

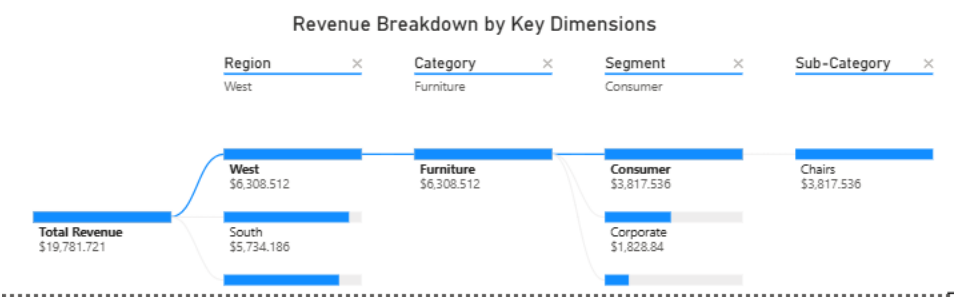
Matrix Visual

Revenue, Profit, and Orders by Category and Region

Region	Central			East
Category	Total Revenue	Total Profit	Total Orders	Total Revenue
Furniture	\$5,276.472	\$425.8922	9	\$2,462.551
Total	\$5,276.472	\$425.8922	9	\$2,462.551

A matrix visual was used to summarize revenue, profit, and order counts across categories and regions. This provides a compact tabular view for comparing performance across multiple dimensions.

Decomposition Tree

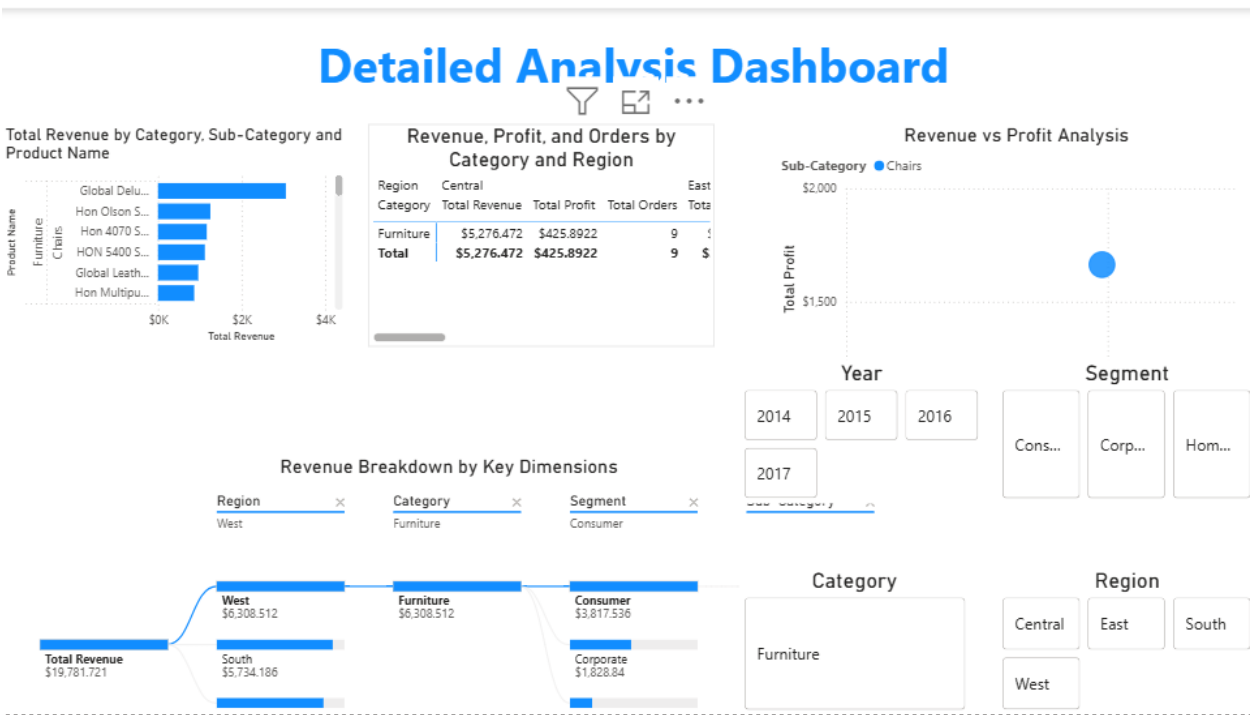


A decomposition tree was used to break down revenue by different dimensions such as region, category, and customer segment. This helps users understand the key drivers of performance.

Advanced Analytical Visual



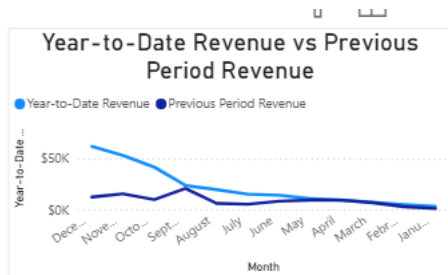
A scatter chart was included as an advanced analytical visual to compare revenue and profit across products or sub-categories. This makes it easier to identify strong performers, weak performers, and unusual patterns.



PAGE 3: Insights and Performance Monitoring

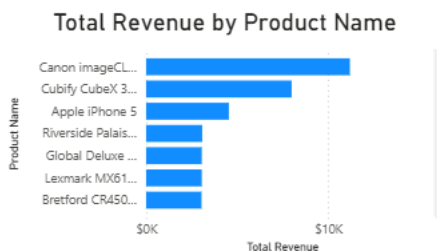
The third report page focuses on monitoring performance over time, identifying top and bottom performers, and comparing current results with previous periods.

Time Intelligence Analysis



A time intelligence visual was added to compare year-to-date revenue with previous period revenue. This helps users monitor current performance against historical performance.

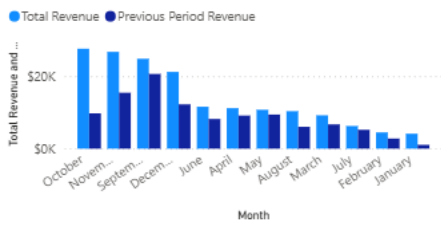
Top Performers Visuals



A top performers chart was created to identify the products generating the highest revenue. This helps management understand which products are driving performance.

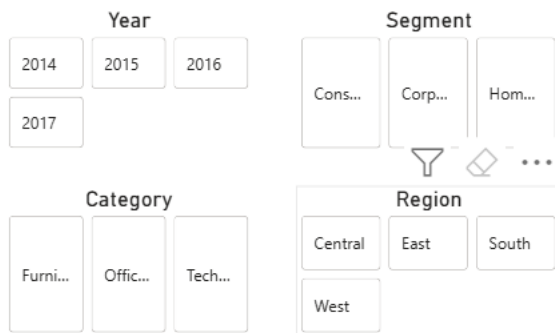
Comparison or Anomaly Analysis

Total Revenue and Previous Period Revenue by Month



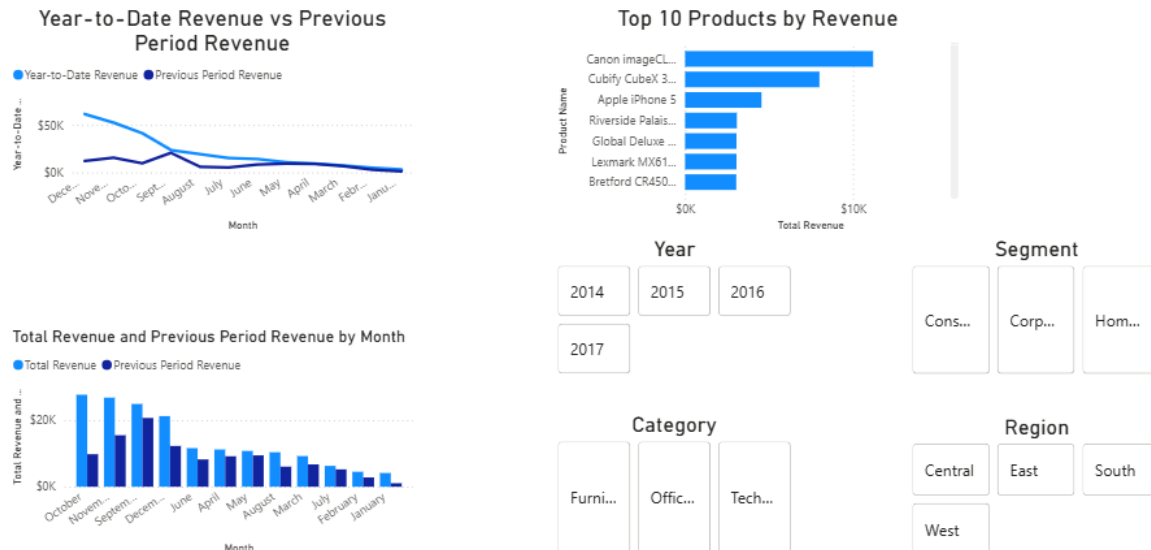
A comparison visual was used to evaluate current performance against previous periods. This helps identify unusual changes, growth trends, and potential anomalies in business performance.

Meaningful Cross-filtering



Cross-filtering was enabled across visuals so that selecting an item in one chart dynamically updates the other visuals on the page. This improves exploration and helps users understand relationships between performance indicators.

Insights and Performance Monitoring



Dashboard Design

The dashboard was designed using a business-friendly layout with clear titles, consistent formatting, and logical grouping of visuals. Summary visuals were placed on the executive page, detailed exploration visuals were placed on the analysis page, and monitoring visuals were placed on the performance page. This structure helps users move from overview to detail in a natural way.

Visual types were selected based on purpose. KPI cards were used for headline figures, line charts were used for trends, bar and column charts were used for comparisons, a matrix was used for tabular summaries, and a decomposition tree and scatter chart were used for deeper analytical exploration.

Interactive slicers and cross-filtering were included to allow users to explore the dashboard by year, region, segment, and category. These features improve usability and help different stakeholders focus on the information most relevant to them.

How users interact with the dashboard

Users interact with the dashboard through slicers, chart selections, and drill-down functionality. Slicers allow filtering by dimensions such as year, category, segment, and region. Drill-down visuals allow users to move from category-level summaries to more detailed product-level insights. Cross-filtering ensures that selections in one visual automatically update other visuals, making the dashboard interactive and easy to explore.

Part F: Analytical Insights and Business Recommendations (10 Marks)

The dashboard provides a comprehensive view of business performance across revenue, profit, customers, products, and time. Using interactive visuals and time intelligence measures, several key patterns, trends, and performance issues were identified. The following section presents analytical insights derived from the dashboard, followed by actionable business recommendations.

Analytical insights

1. Declining Revenue Trend Over Time

The revenue trend shows a noticeable decline from peak months (around October–December) towards later months such as January and February. This indicates a seasonal pattern where sales are strongest towards the end of the year but weaken significantly afterward.

Why this matters:

- Indicates **seasonality**
- Risk of **low performance periods**
- Suggests uneven revenue distribution across the year

2. Technology Category is the Top Revenue Driver

The category comparison chart shows that **Technology consistently generates the highest revenue**, outperforming Furniture and Office Supplies.

Why this matters:

- Technology is the **core revenue contributor**
- Business is **dependent on one category**
- Opportunity to scale or risk if demand drops

3. Revenue Concentration in Few Top Products

The “Top 10 Products by Revenue” chart indicates that a small number of products contribute a disproportionately large share of total revenue.

Why this matters:

- Indicates **high product concentration**
- Business risk if these products underperform
- Suggests **Pareto effect (80/20 rule)**

4. Gap Between Revenue and Profit (Efficiency Issue)

The scatter plot (Revenue vs Profit) shows that not all high-revenue products generate high profit. Some products generate strong sales but relatively lower profit margins.

Why this matters:

- Indicates **inefficient pricing or high costs**
- Some products may not be financially optimal
- Profitability is not aligned with revenue

5. Customer Base is Broad but Possibly Low Value

The dashboard shows a relatively high number of distinct customers (793), but the average order value (~\$212) suggests that individual transactions are not very large.

Why this matters:

- Business relies on **volume rather than high-value transactions**
- Potential to increase revenue through:
 - upselling
 - bundling
 - premium offerings

BUSINESS RECOMMENDATIONS

1. Strengthen Sales During Low-Performance Months

Since revenue declines significantly after peak months, the business should introduce targeted promotions, discounts, and marketing campaigns during low-performing periods to stabilize revenue throughout the year.

Impact:

- Reduces seasonality impact
- Improves revenue consistency
- Better cash flow management

2. Focus on High-Performing Product Categories While Diversifying

The business should continue investing in the Technology category while also improving the performance of underperforming categories such as Furniture and Office Supplies through better pricing, marketing, or product improvements.

Impact:

- Maintains strong revenue drivers
- Reduces dependency risk
- Expands overall revenue base

3. Improve Profitability of High-Revenue Products

Products that generate high revenue but low profit should be reviewed for cost optimization, pricing adjustments, or supplier renegotiation to improve profit margins.

Impact:

- Increases overall profitability
- Improves financial efficiency
- Ensures sustainable growth

Conclusion

Overall, the dashboard reveals key opportunities for improving business performance through better demand management, product strategy optimization, and targeted customer engagement. By addressing seasonal fluctuations, improving profit margins, and leveraging high-performing products and regions, the business can achieve more stable and sustainable growth.

PUBLISHING EVIDENCE

Publishing to Power BI

✓ Success!

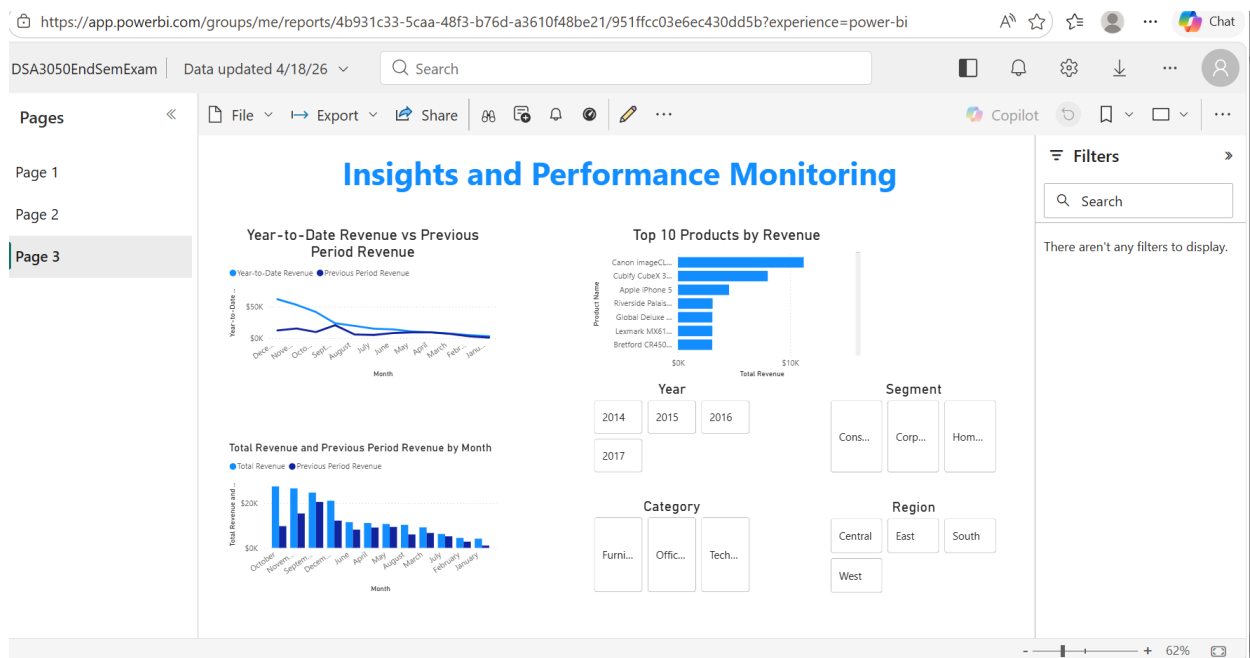
[Open 'DSA3050EndSemExam.pbix' in Power BI](#)

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You can create a portrait view of your report, tailored for mobile phones.
On the **View** tab, select **Mobile Layout**. [Learn more](#)

Got it

Dashboard open in Power BI service



Dashboard link

<https://app.powerbi.com/view?r=eyJrIjoizTQ2N2I2NTYtZWlZNS00NTc3LWEyMWMtMjk2YjgwMmVIYTg2liwidCI6IEZ2DgzZWU2LTI1NGEtNDY5ZC1hNmNjLTU0ZTJjYTlzMtNlNylslmMiOjh9>

GitHub Repository Link

<https://github.com/Zakishafi/DSA3050A-Advanced-PowerBI-Exam-Zakariya-Shafi-596.git>